

Worksheet 4E: Optimisation — Calculus Finds the Best Answer — Solutions

Series 4: Calculus
AIML Foundations Mathematics
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Instructor Copy — Not for Distribution

Note to instructor: Key answers are given below. Accept any mathematically equivalent form unless the question specifies otherwise.

Part A

1. $f'(x) = 2x - 6 = 0 \Rightarrow x = 3$
2. $g'(x) = 6x^2 - 18x + 12 = 6(x - 1)(x - 2) = 0 \Rightarrow x = 1, 2$
3. $h'(x) = 4x^3 - 16x = 4x(x^2 - 4) = 0 \Rightarrow x = 0, \pm 2$
4. $f'(x) = 3x^2 = 0 \Rightarrow x = 0$. f' does not change sign — the function is non-decreasing everywhere (an inflection point).
5. $f'(x) = \frac{2x(x^2+1)-2x(x^2-1)}{(x^2+1)^2} = \frac{4x}{(x^2+1)^2} = 0 \Rightarrow x = 0$
6. $f'(x) = e^x - e = 0 \Rightarrow x = 1$
7. $f'(x) = 1 - 1/x = 0 \Rightarrow x = 1$
8. $f'(x) = \cos x - \sin x = 0 \Rightarrow \tan x = 1 \Rightarrow x = \pi/4, 5\pi/4$
9. $f'(x) = e^{-x} - xe^{-x} = e^{-x}(1 - x) = 0 \Rightarrow x = 1$
10. $f'(x) = \frac{1-\ln x}{x^2} = 0 \Rightarrow \ln x = 1 \Rightarrow x = e$

Part B

11. $f'(x) = 2x - 6$; critical point $x = 3$. $f''(3) = 2 > 0 \Rightarrow$ local min. Min value $f(3) = -4$.
12. $x = 1$: $g''(1) = 12 - 18 = -6 < 0 \Rightarrow$ local max, $g(1) = 1$. $x = 2$: $g''(2) = 12 > 0 \Rightarrow$ local min, $g(2) = 0$.
13. $x = 0$: $h''(0) = -16 < 0 \Rightarrow$ local max. $x = \pm 2$: $h''(\pm 2) = 32 > 0 \Rightarrow$ local mins.
14. $x = 1$: $f''(x) = e^{-x}(x - 2)$; $f''(1) = -e^{-1} < 0 \Rightarrow$ local max. Max value e^{-1} .
15. $x = e$: $f''(e) = -(1/e^3)(1) < 0$ (compute via quotient rule) $\Rightarrow$ local max. Max value $1/e$.
16. Local max at $x = 3$.
17. e.g. $f(x) = x^4$, $g'(0) = g''(0) = 0$; but $x = 0$ is a local min.

18. (Sketch: smooth curve with described features.)
19. $f'(x) = 2ax + b = 0 \Rightarrow x = -b/(2a)$. $f''(x) = 2a$: positive when $a > 0$ (min), negative when $a < 0$ (max).
20. For even n : $f'(x) = nx^{n-1} = 0$ at $x = 0$; $f''(0) = n(n-1) \cdot 0^{n-2} \dots$ use first derivative test: f' is negative for $x < 0$ and positive for $x > 0 \Rightarrow$ local min. For odd n : no sign change $\Rightarrow$ inflection point.

Part C

21. (a) Fencing: $2x + y = 120$, $y = 120 - 2x$. $A(x) = x(120 - 2x) = 120x - 2x^2$. (b) $A'(x) = 120 - 4x = 0 \Rightarrow x = 30$ m, $y = 60$ m. (c) $A = 1800$ m².
22. (b) $h = 32/x^2$; $C = 2x^2 + 4xh = 2x^2 + 128/x$. (c) $C' = 4x - 128/x^2 = 0 \Rightarrow x^3 = 32 \Rightarrow x = 2\sqrt[3]{4} \approx 3.17$ m.
23. (a) $h = 500/(\pi r^2)$; $S(r) = 2\pi r^2 + 1000/r$. (b) $S'(r) = 4\pi r - 1000/r^2 = 0 \Rightarrow r^3 = 1000/(4\pi) \Rightarrow r \approx 4.30$ cm; $h \approx 8.60$ cm. (c) $h/r = 2$.
24. (a) $P(x) = 40x - 0.5x^2 - 200$. (b) $P'(x) = 40 - x = 0 \Rightarrow x = 40$. (c) $P(40) = 800 - 800 - 200 \dots P(40) = 40(40) - 0.5(1600) - 200 = 1600 - 800 - 200 = 600$.
25. Let landing point be x km along coast from nearest point. Time = $\frac{\sqrt{16+x^2}}{20} + \frac{9-x}{30}$. Differentiate; set to 0: $\frac{x}{20\sqrt{16+x^2}} = \frac{1}{30}$. Solve: $30x = 20\sqrt{16+x^2}$; $9x^2 = 4(16+x^2)$; $5x^2 = 64$; $x = 8/\sqrt{5} \approx 3.58$ km.
26. Let width = w , height = h ; semicircle radius = $w/2$. Perimeter: $2h + w + \pi w/2 = 10$. Area = $wh + \pi(w/2)^2/2$. Eliminate h , differentiate: $w = 10/(2 + \pi/2) \approx 2.8$ m.
27. (a) $C'(x) = 0.12x^2 - 1.8x + 10$ (b) $C''(x) = 0.24x - 1.8 = 0 \Rightarrow x = 7.5$ (c) Min avg cost where $C' = \bar{C}$: $0.12x^2 - 1.8x + 10 = 0.04x^2 - 0.9x + 10 + 200/x$; solve numerically or set $\bar{C}' = 0$.
28. (a) $L(m) = \frac{1}{3}(14m^2 - 52m + 38)$ (b) $\frac{dL}{dm} = \frac{1}{3}(28m - 52) = 0 \Rightarrow m = 52/28 = 13/7 \approx 1.857$ (c) $m = 13/7$.
29. $L(p) = -\ln(p)$; $L'(p) = -1/p$, never zero. Min is at $p \rightarrow 1$ (boundary of domain): $L(1) = 0$.
30. Total time $T = \sqrt{d_1^2 + x^2}/v_1 + (D - x)/v_2$ where d_1 is perpendicular distance. $dT/dx = x/(v_1\sqrt{d_1^2 + x^2}) - 1/v_2 = 0$; since $x/\sqrt{d_1^2 + x^2} = \sin \theta_1$, we get $\sin \theta_1/v_1 = 1/v_2 = \sin 90^\circ/v_2$, i.e. $\sin \theta_1/v_1 = \sin \theta_2/v_2$.
31. $L_{\text{reg}}(m) = \frac{14}{3}m^2 - \frac{52}{3}m + \frac{38}{3} + \lambda m^2$; optimal $m = \frac{52/3}{28/3 + 2\lambda} = \frac{52}{28 + 6\lambda}$. Increasing λ shrinks m toward 0.
32. $A = 4xy$ subject to $x^2 + y^2 = 25$. By symmetry or Lagrange: $x = y = 5/\sqrt{2}$; square with side $5\sqrt{2}$; area = 50.

Part D

33. $f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$. Evaluate: $f(-2) = -2$, $f(-1) = 4$, $f(1) = 0$, $f(2) = 4$. Absolute max: 4 at $x = -1$ and $x = 2$; absolute min: -2 at $x = -2$.
34. $g'(x) = 1 + 4/x^2 > 0$ always — no critical points. Evaluate endpoints: $g(1) = -3$, $g(4) = 3$. Abs min -3 , abs max 3 .
35. $h'(x) = 2 \cos x - 2 \sin 2x = 2 \cos x(1 - 2 \sin x) = 0$: $x = \pi/2$, $x = \pi/6$, $x = 5\pi/6$. Evaluate: $h(0) = 1$, $h(\pi/6) = 3/2$, $h(\pi/2) = 1$, $h(5\pi/6) = 3/2$, $h(\pi) = -1$. Max: $3/2$; min: -1 .
36. (a) $T'(t) = -2t + 6 = 0 \Rightarrow t = 3$ h. (b) Max: $T(3) = 19^\circ\text{C}$; min: $T(0) = 10^\circ\text{C}$ or $T(8) = 10 - 48 + 10 = -28^\circ\text{C}$ — check: $T(8) = -64 + 48 + 10 = -6^\circ\text{C}$. Min is -6°C at $t = 8$.
37. Let sides be x and $P/2 - x$ (perimeter $= 2(x + (P/2 - x)) = P$). Area $= x(P/2 - x)$; differentiate to get $x = P/4$, i.e. a square.
38. $(0, 1)$ is open; $1/x \rightarrow \infty$ as $x \rightarrow 0^+$ — no maximum. The EVT requires a closed interval.

Part E

39. $x_1 = 4 - 0.1(8) = 3.2$; $x_2 = 3.2 - 0.64 = 2.56$; $x_3 \approx 2.048$; $x_4 \approx 1.638$. Approaching 0.
40. $x_1 = 4 - 0.5(8) = 0$. Converges in one step!
41. $x_1 = 4 - 1.1(8) = -4.8$; $x_2 = -4.8 - 1.1(-9.6) = 5.76$; oscillates and diverges. Too large α causes overshooting.
42. $f'(x) = 2(x - 3)$. Steps: $x_0 = 0$, $x_1 = 0 - 0.2(-6) = 1.2$, $x_2 = 1.2 - 0.2(-3.6) = 1.92$, $x_3 \approx 2.352$, $x_4 \approx 2.611$, $x_5 \approx 2.769$. Converging to 3.
43. $dL/dm = (28m - 52)/3$. $m_1 = 0 - 0.1(-52/3) \approx 1.733$; $m_2 \approx 1.733 - 0.1(28(1.733) - 52)/3 \approx \dots$; converging toward $13/7 \approx 1.857$.
44. (a) Very slow convergence, many iterations needed. (b) Overshooting — oscillates or diverges. (c) (Sketch showing three characteristic curves.)
45. A mini-batch gradient is an unbiased estimate of the true gradient. Averaged over many steps, the noise cancels out and descent still progresses toward the minimum.
46. (a) Each step reduces f if the step is small enough; we stop only when $f'(x_n) \approx 0$. (b) For a convex function, every local min is global; gradient always points toward the unique minimum. (c) $\mathbf{w} \leftarrow \mathbf{w} - \alpha \nabla_{\mathbf{w}} L$, where $\nabla_{\mathbf{w}} L = \frac{\partial L}{\partial \mathbf{w}}$ is the Jacobian.